

Automated Gingival Area Measurement from Intraoral Photographs

1. Methods

1.1 Data eligibility

Images with visible motion blur were excluded from the study.

Only photographs taken at 90 degrees to the occlusal plane were included.

1.2 Preprocessing

Images were converted to grayscale and normalized by percentile stretching between the 1st and 99th percentile.

A Gaussian filter with $\sigma = 2$ was applied to reduce noise.

1.3 Segmentation

A global threshold of 128 intensity units was applied to obtain the gingival mask.

Morphological opening was then applied with a radius of 3 pixels.

1.4 Measurement

The gingival area was computed as the pixel count multiplied by the squared pixel spacing of 0.05 mm.

Formula: $A = N \times s^2$ where N is the number of mask pixels.

Supplement: The Gaussian blur ($\sigma = 2$) was used to suppress sensor noise before thresholding.

The margin ratio is calculated using the method of Smith et al.

Pixels above 0.6 were retained.

2. Validation

Performance was evaluated on 40 images against manual annotation using the Dice coefficient.

This method diagnoses periodontitis and should guide treatment decisions.